[11]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

iris=load_iris(as_frame=True)
x=iris.data
print("dataset shape",x.shape)

x_scaled=StandardScaler().fit_transform(x)
kms=KMeans(n_clusters=3,n_init=10,random_state=42)
clusters=kms.fit_predict(x_scaled)

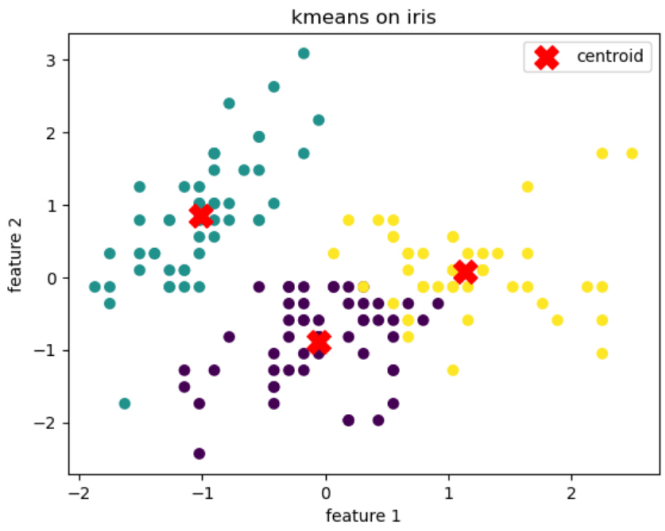
df=x.copy()
df['Cluster']=clusters
print("sample data space\n",df.head())

plt.scatter(x_scaled[:,0],x_scaled[:,1],c=clusters,cmap='viridis')
plt.scatter(kms.cluster_centers_[:,0],kms.cluster_centers_[:,1],color='red',label='centroid',marker='X',s=200)
plt.title("kmeans on iris")
plt.xlabel("feature 1")
plt.ylabel("feature 2")
plt.legend()
plt.show()
```

```
dataset shape (150, 4)
sample data space
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   Cluster
0        1
1        1
2        1
3        1
4        1
```

```
C:\Users\hp\anaconda3\New folder\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```



[12]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

[13]:
```
iris=load_iris(as_frame=True)
x=iris.data
print("dataset shape",x.shape)
```

```
dataset shape (150, 4)
```

[14]:
```
x_scaled=StandardScaler().fit_transform(x)
kms=KMeans(n_clusters=3,n_init=10,random_state=42)
clusters=kms.fit_predict(x_scaled)
```

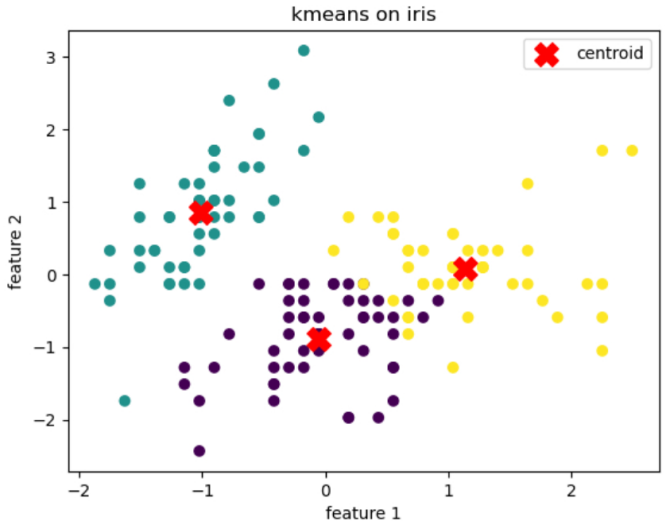
```
C:\Users\hp\anaconda3\New folder\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MK
L, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

[15]:
```
df=x.copy()
df['Cluster']=clusters
print("sample data space\n",df.head())
```

```
sample data space
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   Cluster
0        1
1        1
2        1
3        1
4        1
```

[16]:
```
plt.scatter(x_scaled[:,0],x_scaled[:,1],c=clusters,cmap='viridis')
plt.scatter(kms.cluster_centers_[:,0],kms.cluster_centers_[:,1],color='red',label='centroid',marker='X',s=200)
plt.title("kmeans on iris")
plt.xlabel("feature 1")
plt.ylabel("feature 2")
plt.legend()
plt.show()
```



[ ]: